

Joseph Peters

Summary of Qualifications

Highly motivated and results driven high school student with a proven record of leadership, technical innovation, and community impact. Possesses advanced skills in programming (Python, Java, CAD) and a passion for applying engineering principles to solve complex social and technical challenges. Seeking to leverage pre college academic and professional experiences in a rigorous university environment.

Education

Johns Hopkins University *Engineering Innovation Summer Program* | Summer 2025

- **Achieved** a final grade of A in this intensive pre college course, earning college credit.
- **Mastered** core Engineering Principles (civil, chemical, electrical/computer, mechanical) through real world applications and problem solving.
- **Applied** analytical tools and techniques to design, construct, and test structures, including participation in a competitive bridge building challenge.
- **Completed** hands on and virtual experiments, culminating in oral presentations on product design.

Montgomery College *Intermediate Java Programming* | August 2024

- **Completed** a 24 hour course through the Workforce Development and Continuing Education program.
- **Gained** proficiency in Intermediate Java Programming, expanding technical skill set.

St. Andrew's Episcopal School *Potomac, MD* | Expected Graduation: [Year]

Experience & Technical Projects

Nika *Technical Intern | Summer 2024*

- **Completed** a one month technical internship at a private government firm, receiving direct mentorship from professional engineers.

HackBak *Engineering & UI/UX Design Lead | 2023 Present*

- **Engineered** a winning prototype for the 2025 HackBak competition: “DiscoverU,” an educational game linking job opportunities with user passions.
- **Developed** two distinct social impact prototypes addressing critical issues: one focused on solving depression and weight problems for people of color, and another targeting gentrification in underprivileged communities.

American University Development Project *Project Lead | Summer 2022*

- **Conceptualized** and **launched** “Achieve Balance,” a comprehensive scheduling solution for high school athletes struggling with academics, sports, and mental health.
- **Designed** a tangible and digital tool that successfully integrates goal setting, organization, and self care into daily routines.

Rover Sorter *Independent Passion Project | In Progress*

- **Designed** and **built** a functional rover with an articulated arm capable of sorting plastics, metals, and recyclable materials.
- **Programmed** the entire system using a Raspberry Pi and Python.

Leadership & Activities

Robotics Club *Programming and Design Lead | 2022 Present*

- **Orchestrated** programming and design efforts as a core member, directly contributing to the development of competition ready robots.
- **Designed** and **manufactured** key robot components using CAD software and hands on fabrication techniques.

- **Competed** in multiple robotic tournaments, demonstrating technical expertise and collaborative problem solving under pressure.
- **Mentored** team members on coding, engineering, and project management best practices.

Leukemia & Lymphoma Society (LLS) *Top Fundraiser*

- **Spearheaded** fundraising efforts that **generated \$50,000** for the LLS chapter.
- **Recognized** as the Second Runner Up for DC Student Visionary Team Member of the Year (ranking 3rd out of 400 total students in the campaign).

Chess Club *Co Captain | 2022 Present*

- **Co led** the club and **prototyped** an application to facilitate inter school chess club play and community building.

Set Design *Management Staff Member | 2023 and 2025*

- **Managed** and **executed** the construction of sets for the school's theatre productions.
- **Contributed** to the management staff responsible for directing the school plays.

Research

- **Conducting** a research paper on the Gamification of Education (in progress).

Skills

- **Programming Languages:** Python, Java
- **Technical Tools:** CAD, Raspberry Pi
- **Interests:** Piano